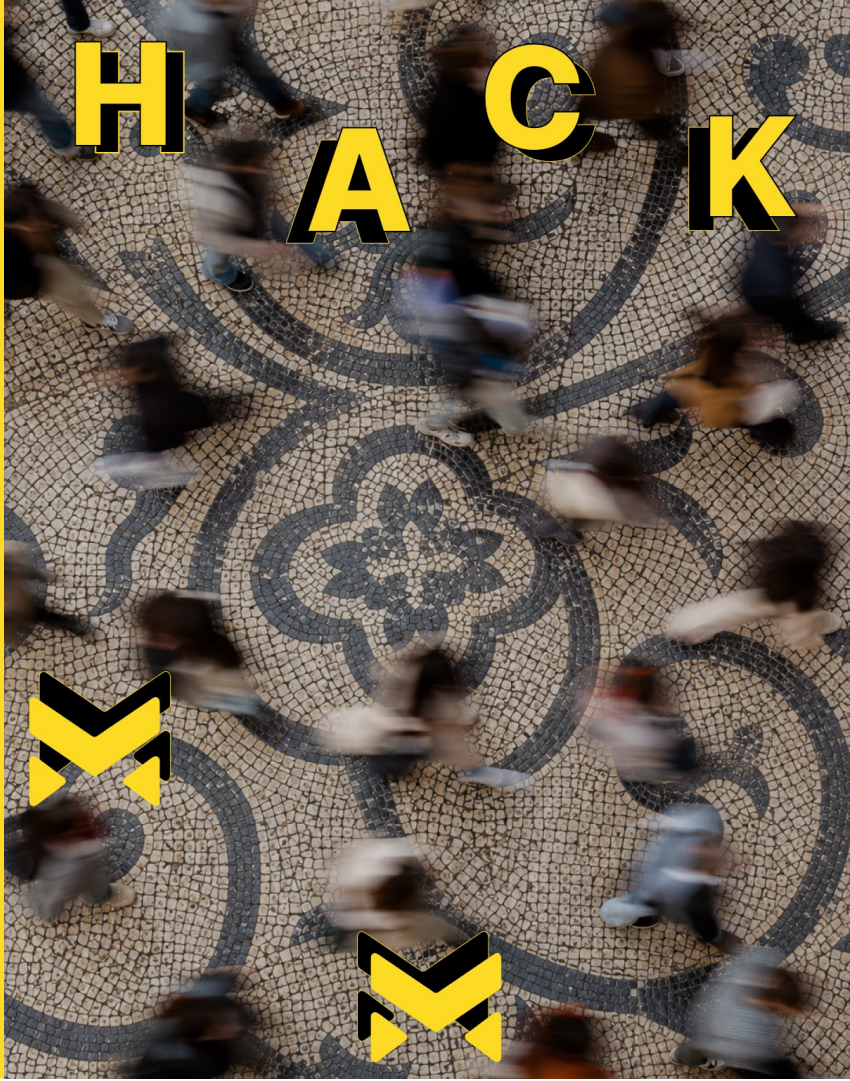
Hack Meridian / 2026 LISBON

BUILDING AT HACK MERIDIAN

STELLAR SMART ACCOUNTS

Why a C-address, and how to build an app on one

Stellar





YOUR ACCOUNT SHOULD HAVE RULES, NOT JUST A KEY

G-addresses hold keys. C-addresses run code.

CLASSIC ACCOUNTS

G-ADDRESS: A KEY

Ed25519 signers, weights, thresholds:
fixed rules the protocol defines.
An app asks every time, or holds your key.

Multisig exists. Scoped permission does not.





CONTRACT ACCOUNTS

C-ADDRESS: AN ACCOUNT THAT RUNS CODE

from.require_auth()
→ the account's __check_auth
→ its rules decide

Same Address type for your contract

WHAT RULES BUY YOU

1

PASSKEYS
Face ID instead of a seed phrase

2

SCOPED KEYS
One contract, listed functions, an expiry

3

SPENDING CAPS
A rolling limit the account enforces

THE HONEST TRADE-OFFS

WHAT NIDO'S RELAYER PAYS

Account: about 1.4 XLM
First grant: about 4.5 XLM
Capped grant: about 2.7 XLM

From one 100 XLM/day pot, on 22 Oct 2026

COMPATIBILITY

No classic operations; Horizon shows no account

COST

Sponsored, up to a shared daily budget

MATURITY

Testnet, pre-audit, forked dependencies

PERCH: A RULE YOU CAN READ

{ "name": "kudos-xlm",
  "scope": { "type": "contract", "address": "CDLZ...CYSC" },
  "principals": { "type": "all", "signers": ["session"] },
  "functions": ["transfer"],
  "cap": { "limit": "20000000", "period-ledgers": 17280 },
  "not-after-ledger": 5342393 }

One document per account, hashed and compiled on chain, enforced in __check_auth.

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THE WALLET

NIDO

Passkey-native Perch accounts.
Connect, sign, or grant a session.
Creation and fees sponsored.

nido.fyi · TESTNET



THE POPUP CAN'T SIGN FOR YOUR CONTRACT

```
guess(3, CAXS...) → __check_auth
  context_rule_ids: [0]  // "admin", scoped to the account
  contexts: guess, transfer  // two calls, one rule
→ Error(Contract, #3014) ContextRuleIdsLengthMismatch
```

One contract: #3002. Several: #3014. Fix: ask once for a session.



ONE PROMPT: KUDOS JAR

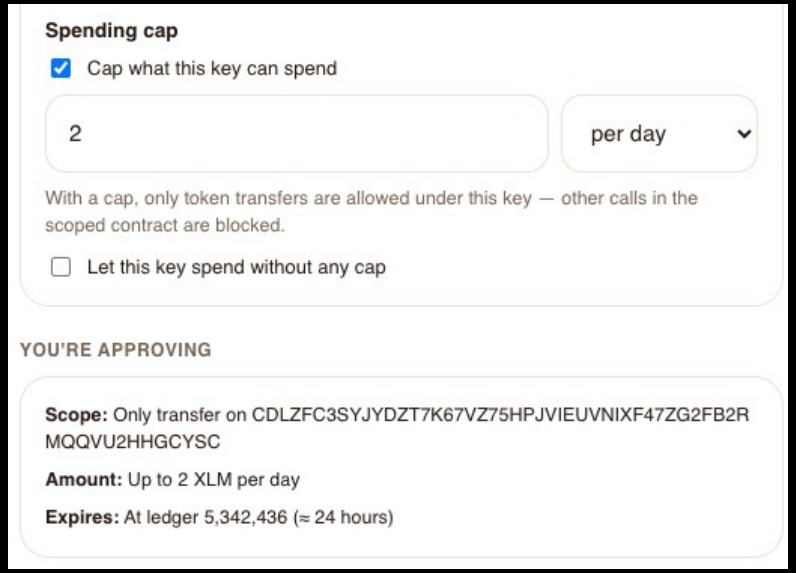
01 Copy the template with new-app

02 Write and test the contract

03 Deploy it to testnet

04 Run it with a real Nido account

APPROVE THE RULE IN NIDO



Nido's grant page, before the passkey

SCOPE

Only transfer, only on XLM

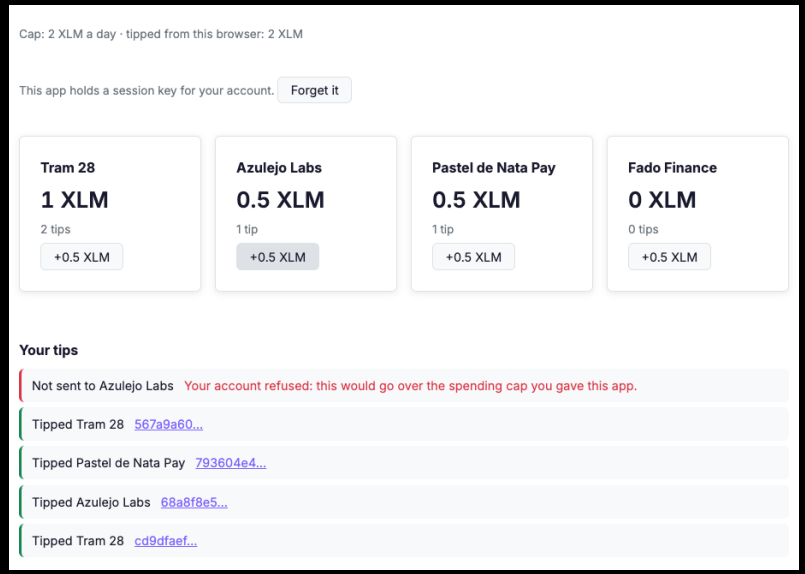
CAP

Up to 2 XLM per day

EXPIRY

About 24 hours, then gone

TAP, TAP, TAP... REFUSED



Live testnet, 22 Oct 2026

NO PROMPTS

Four tips, each confirmed

THE CAP

The fifth would pass 2 XLM

ON CHAIN

The account said no: #3221

WHEN THE WALLET SAYS “UNKNOWN”

CHECK THE CHAIN BEFORE YOU RETRY

Go back to the app, press Check again: it reads the account’s rules on chain. Nothing there? Ask Nido again.

A failed response can follow a success

429 · BUDGET USED UP

Nothing was sent. Wait, or pay your own fee

503 · RELAYER BUSY

Nothing was sent. Retry in a moment

504 · TIMED OUT

It may have landed. Check, then retry



PATTERNS THAT WORKED

01 Per-user cap: a rule on the token

02 Shared pot: a contract, 2 of 3 approve

03 No money moves: one grant, no cap

04 Allowance or time lock: a vault holds it

YOUR OWN APP, START TO FINISH

```
npm run new-app -- ../my-app --blank && cd ../my-app
npm run tools:scaffold && npm ci && npm run doctor
# grow contracts/check-in, or add a crate and list it in scaffold.yml
npm run deploy:testnet  # deploys, pins its testnet ID, writes the client
<SessionGate grants={grants}>  // one Grant per contract in env.auths()
const { result, hash } = await submitCall(address, (publicKey) ⇒
  myContract.my_fn({ user: address }, { publicKey }))
```

Helpers from the template’s app-lib. NIDO.md has the whole path; AGENTS.md is for your agent.

KEEP BUILDING

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YOUR FIRST SMART ACCOUNT APP

Give the app a rule, not your key.
Let the account say no.

nido.fyi · testnet
github.com/nidohq/nido
github.com/stellar-registry/perch





BACKUP SLIDES

Perch in depth, the full walkthrough, and what broke

ONE ADDRESS, MANY KEYS

1

RECOVERY
Guardians or a recovery kit

2

SPONSORED START
A relayer pays the fees

3

ADD AND REMOVE
Devices and app keys come and go



POLICY DOCUMENTS

ONE DOCUMENT, ONE HASH

Signers, rules, optional recovery.
Canonical JSON, hashed on chain.
apply_doc is the only way to change it.

Every rule names who, where, what, until when, how much



DOCUMENT TO ENFORCEMENT

01 Write the document

02 Compile it on chain

03 Install as account rules

04 Enforce in __check_auth

VERIFIERS

Coming: ML-DSA (post-quantum) as one more verifier contract. On Perch's roadmap; not built yet.

1

ED25519
Classic keys, as a verifier contract

2

WEBAUTHN
Passkeys on P-256

3

DELEGATED
Another account signs (CAP-71)

CONTENT-ADDRESSED PUBLISHING

CONSTRUCTORLESS

Shared contracts deploy with no arguments.
Address = registry + hash of the wasm.
Accounts pin them by hash; no one repoints.

Upgrades wait ~7 days





ACCOUNTS

ACCOUNTS AND FACTORIES

Rule 0, admin: full owner authority.
Perch factory: the address commits to its signers.
Nido factory: the address exists before the passkey.

Everything else arrives as a document



WHAT AN APP CAN ASK

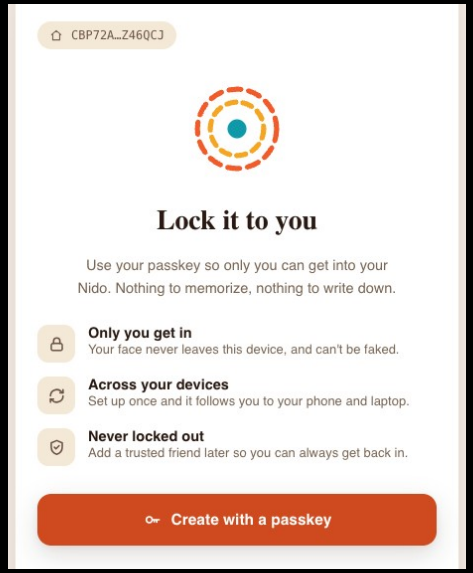
01 Connect: the C-address

02 Sign one transaction

03 Grant a scoped session

04 Recovery stays in Nido

A PASSKEY ACCOUNT



nido.fyi on testnet

RESERVE

The C-address exists first

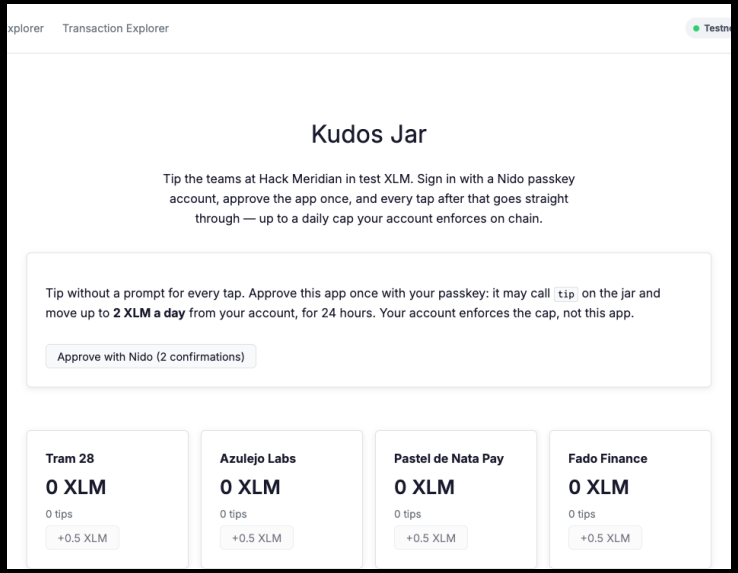
LOCK

One passkey ceremony

FUNDED

5 test XLM, fees sponsored

CONNECT, THEN ASK ONCE



Kudos Jar, running locally

PICK NIDO

From the usual wallet list

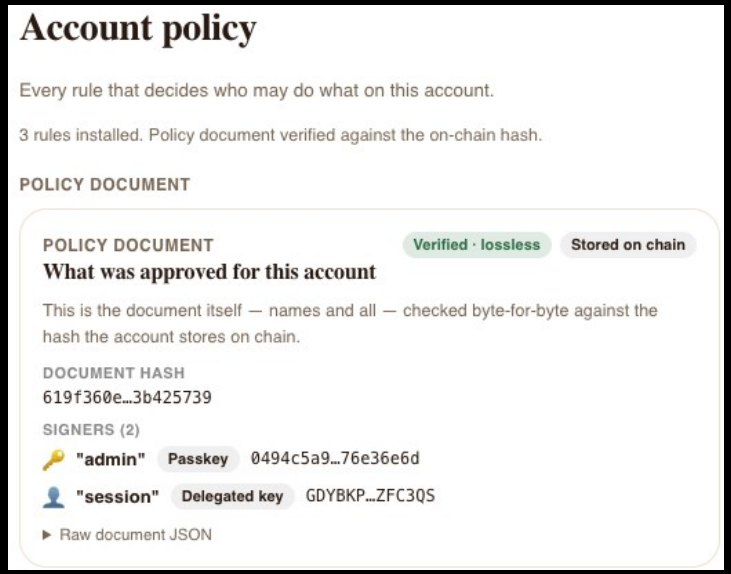
THE ACCOUNT

C-address and its XLM balance

ONE REQUEST

Two rules, two confirmations

THE ACCOUNT'S POLICY



Nido's policy page

DOCUMENT

Hash verified against the chain

RULES

admin, kudos-tip, kudos-xlm

KEYS

A passkey and the app's key

THE CODE THAT MATTERS

```
const tx = await kudosJar.tip(
  { from: address, team: team.id, amount: 5_000_000n },
  { publicKey: transactionSource(address) },
)
const { result } = await submitContract(tx, address)
```

transactionSource and submitContract are template helpers from @stellar-scaffold/app-lib.

WHAT THE AGENT TRIPPED ON

1

SILENT TOOLS
Old Scaffold wrote no clients

2

BLOCKED POPUP
Simulation outlived the click

3

WRONG PATH
Popup signing, then non-root auth

FOUND ALONG THE WAY

1

NIDO SIGNER
Admin rule only: no app calls

2

NIDO RELAYER
Can’t relay CAP-71 entries yet

3

DOCS
Still describe signed XDR



RUN KUDOS JAR

01 npm ci

02 npm run clients:testnet

03 npm run dev

04 Create a Nido, approve, tip